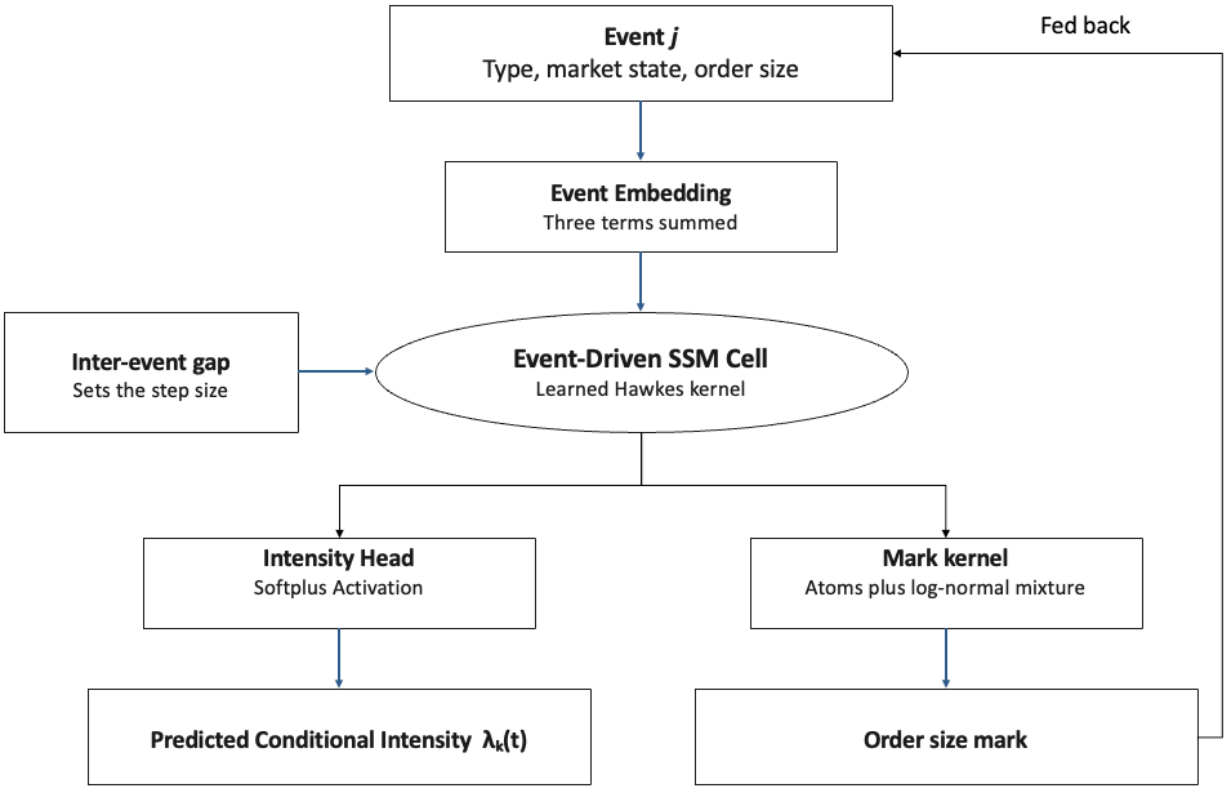
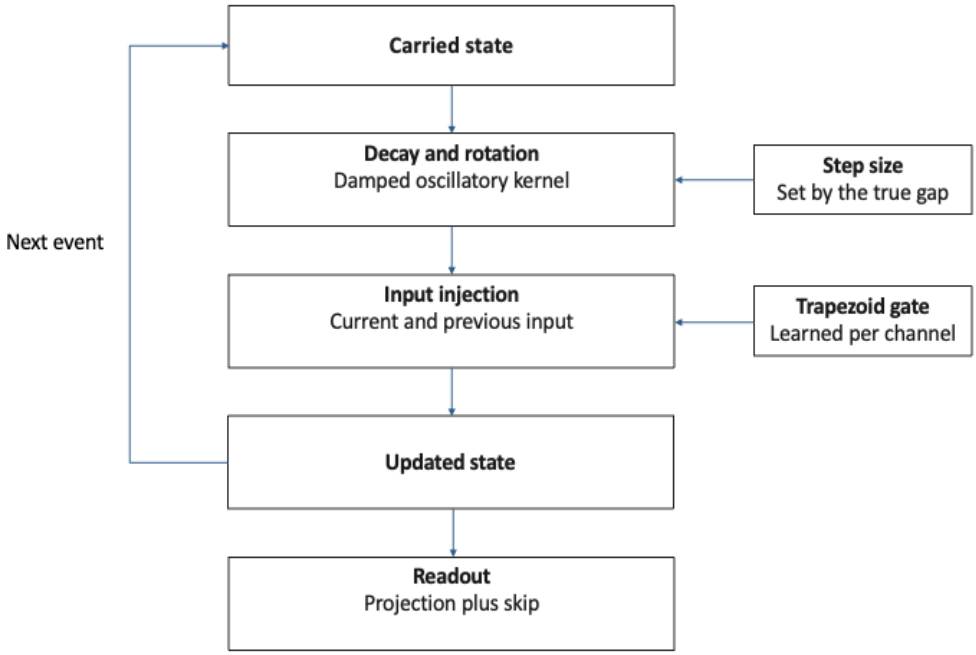
- For a 12-dimensional counting process:
  - $N(t) = \left(N_1, (t), \ldots N_{12}(t)\right)$
- Conditional intensity
  - $\lambda_i(t) \;=\; \phi_i \left(\bar{\lambda}_i + \int_{(0,t)} \sum_{j=1}^{m} \alpha_{ij}\, e^{-\beta_{ij}(t-s)}\, dN_j(s)\right)$
  - $\lambda_k\left(t_j + u \mid \mathcal{H}_{t_j}\right) \;=\; \text{softplus}\left(\alpha_k\, u + w_k^{\top} h_j + b_k\right) \;+\; \mu_k\left(t_j + u\right)$
- Bathtub baseline
  - $\mu_k(t) \;=\; \text{softplus}(g_k) \;+\; \text{softplus}(n_k^0)\, e^{-\upsilon_l \tau} \;+\; \text{softplus}(n_k^T)\, e^{-\upsilon_r (T-\tau)}$
- Order size as a mark
  - $\lambda(t, k, m) \;=\; \lambda_k(t) \cdot f_k(m \mid h)$
  - $f_k(m \mid h) \;=\; \sum_{i=1}^{N} p_{ik}\,(h)\; \mathbf{1}\{m = a_i\} \;+\; p_{0k}(h)\; g_k(\log m \mid h),\; g_k = \sum_{c=1}^{5} w_{ck}\; \mathcal{N}\left(\log m\,; \mu_{ck}; (h)\sigma_{ck}(h)\right)/m$

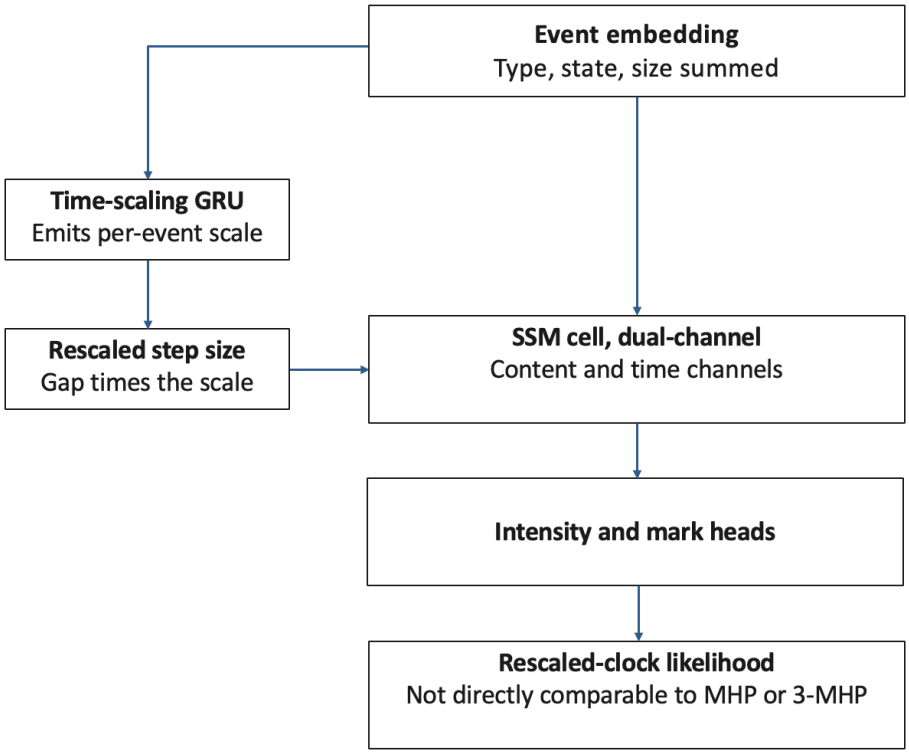
Mamba Hawke Process model:



Mamba-3 Backbone



A-MHP Variant



- Density fit
  - $\ell \;=\; \sum_j \log \lambda_{k_j}\,(t_j) \;-\; \int_{t_0}^{t_J} \Lambda\,(s)\,ds,\; \int_{t_j}^{t_{j+1}} \Lambda\,(s)\,ds \;\approx\; \Delta_{j+1} \cdot \frac{1}{M} \sum_{r=1}^{M} \Lambda\,\left(t_j + u_r\right)$
  - $\mathcal{L} \;=\; -\ell \;+\; w_{sz}\,\mathrm{NLL}_{\mathrm{size}} \;+\; \beta\,\mathrm{CE}\big(k_{j+1}\big) \;+\; \gamma\,\mathrm{MSE}\big(\Delta_{j+1}\big)$
- Point prediction
  - $\mathbb{E}[\Delta t] = \int_0^{\infty} S\,(u)\,du$
  - $S(u) = \exp\big(-\int_0^{u} \Lambda\big)$
- Simulation
  - $\mathbb{P}(i \mid \text{event at } t) \;=\; \frac{\lambda_i(t)}{\Lambda(t)}$,
  - $\Lambda(t) = \sum_{i=1}^{12} \lambda_i\,(t)$
- Midprice construction
  - $V(t + \Delta t) \;=\; V(t) + \frac{\Delta}{2}\left[\sum_{k \in O_u} I_k\,(t)\; a(X_k) \;-\; \sum_{l \in O_d} I_l\,(t)\; b(X_l)\right]$
  - $\mathbb{P}(\Delta |V|(t) = x) = p_i \text{ for } x = j_i,\; i = 1, \ldots, n$

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